

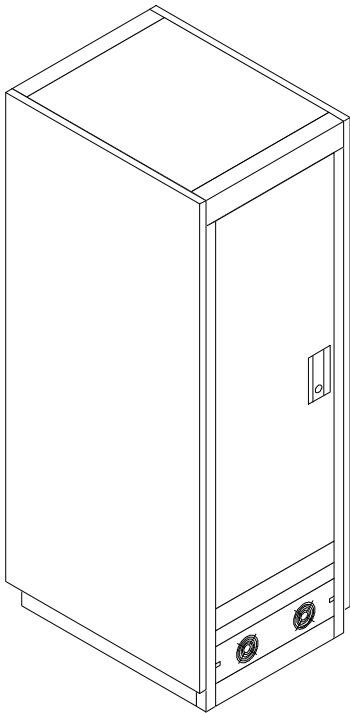
SECTION 5 -- SCHEMATICS

5--1 LIGHTNING POWER DISTRIBUTION UNIT

PD1

LIGHTNING POWER DISTRIBUTION UNIT

46- -317099P20 and P21



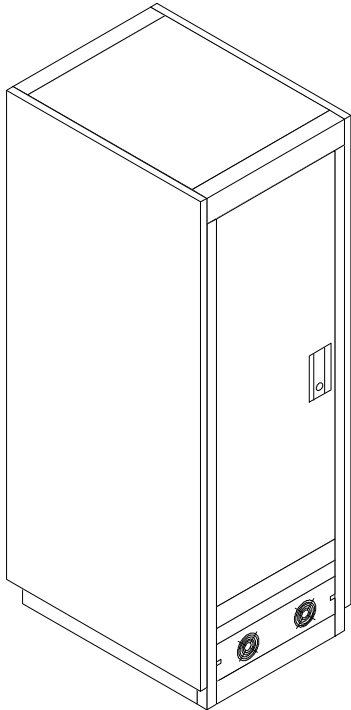
Description

The Lightning Power Distribution Unit (PDU) serves as the primary interface to facility power. It converts the 3Ø input line to either 208VAC (3Ø) or 120VAC (1Ø) as required by each subsystem cabinet. It also provides protection against high voltage spikes and high frequency noise. Additionally, the PDU provides protection against powerline surges and sags and active powerline regulation. The Lightning PDU does **not** provide powerline distortion or frequency shift protection.

5--2 OUTPUT WIRING DIAGRAM

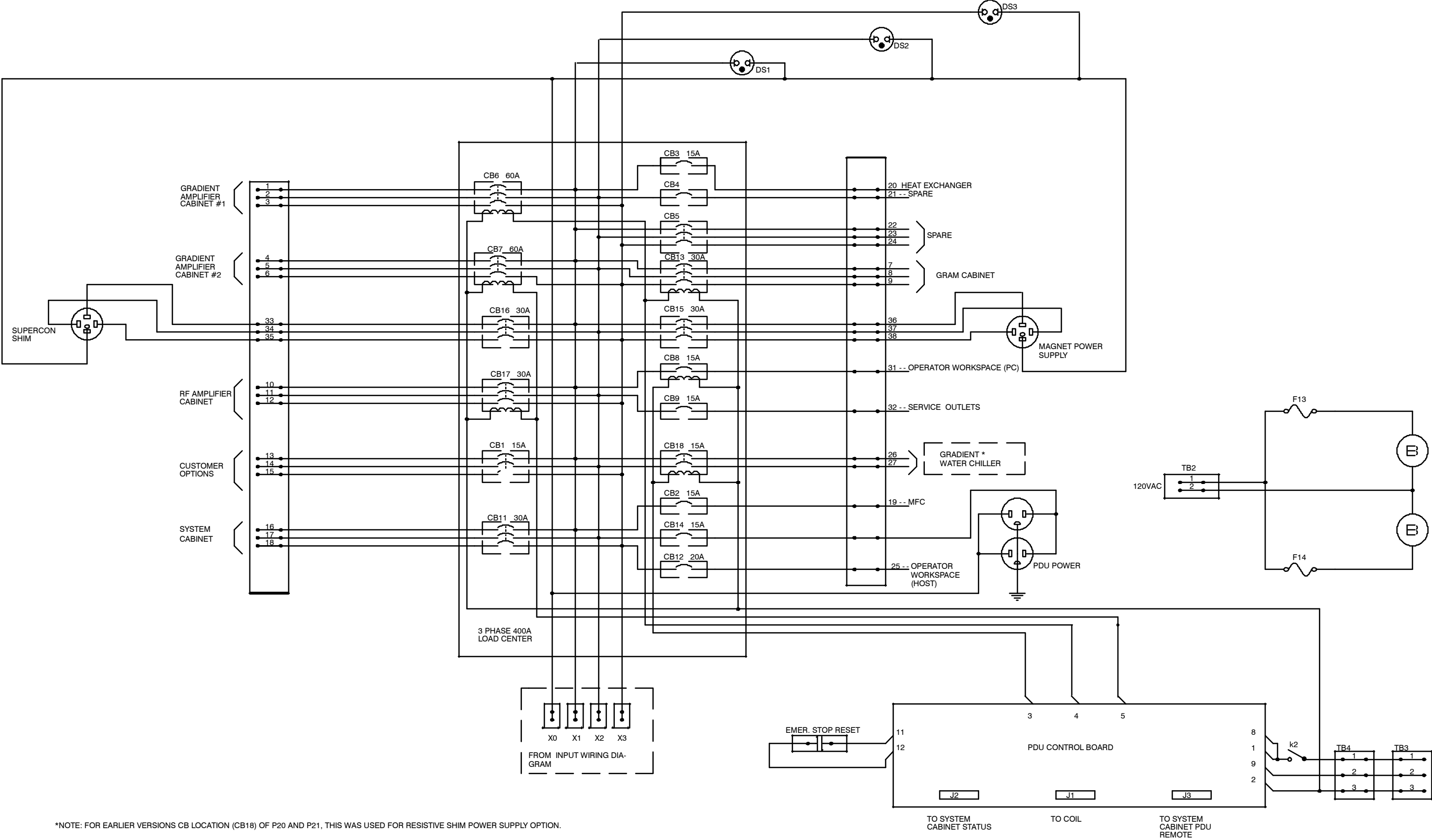
LIGHTNING POWER DISTRIBUTION UNIT

46- -317099P20 AND P21



Description

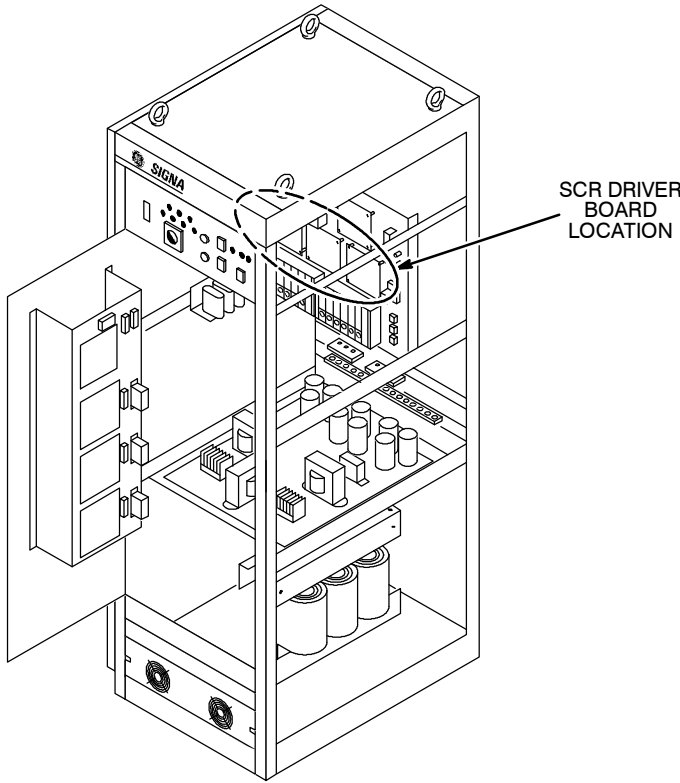
The Lightning Power Distribution Unit (PDU) serves as the primary interface to facility power. It converts the 3 ϕ input line to either 208VAC (3 ϕ) or 120VAC (1 ϕ) as required by each subsystem cabinet. It also provides protection against high voltage spikes and high frequency noise. The Lightning PDU does **not** provide powerline distortion or frequency shift protection.



*NOTE: FOR EARLIER VERSIONS CB LOCATION (CB18) OF P20 AND P21, THIS WAS USED FOR RESISTIVE SHIM POWER SUPPLY OPTION.

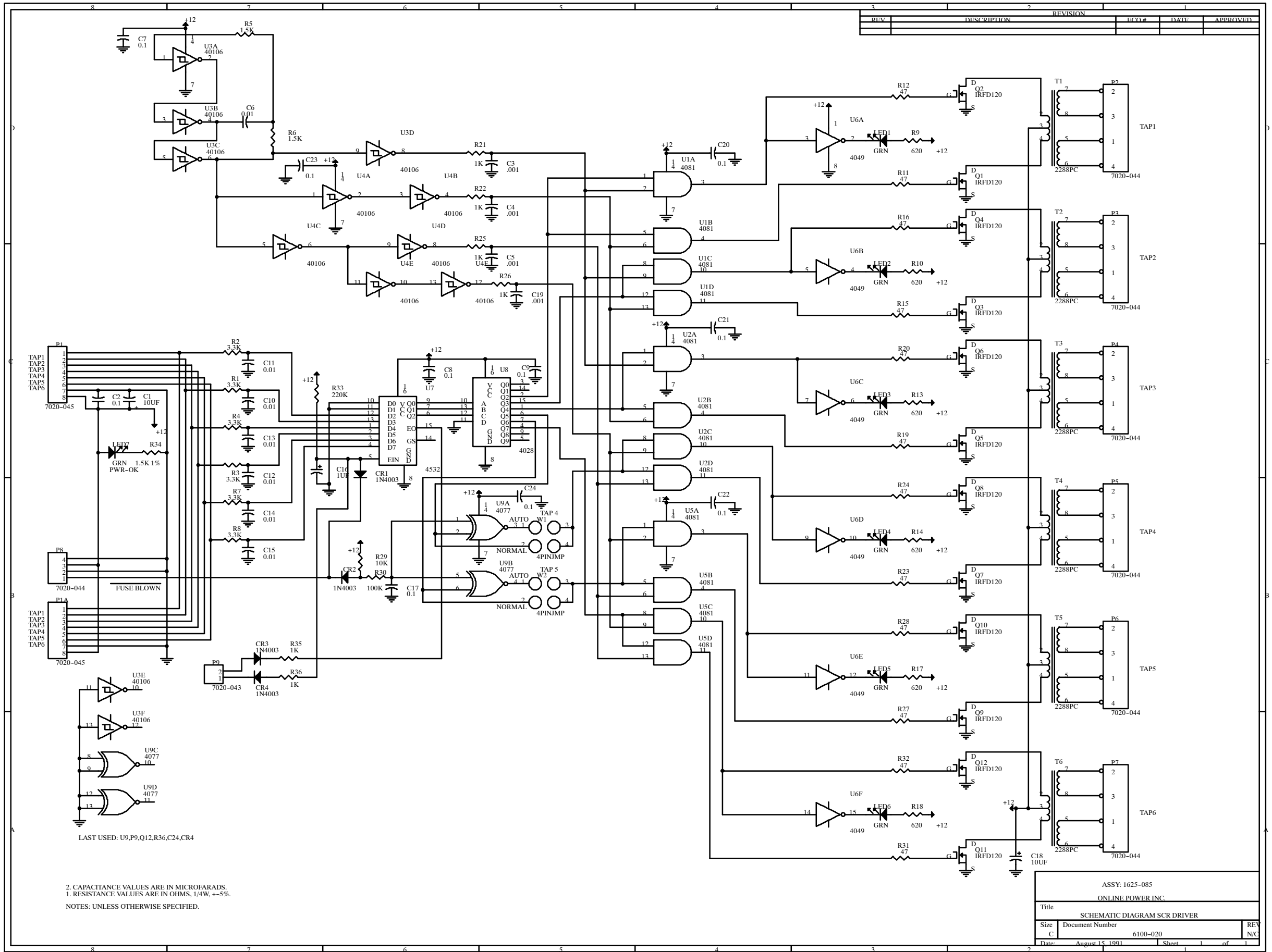
5--4 SCR DRIVER BOARD

PD1 A9 A3, A4, A5
SCR DRIVER BOARD
46- -307819P53



Description

The SCR driver board has a 5 KHz oscillator that provides a push/pull signal to MOSFETs that energize the primary of the pulse transformer. The pulse transformer output is applied to the SCR bank through snubber boards.



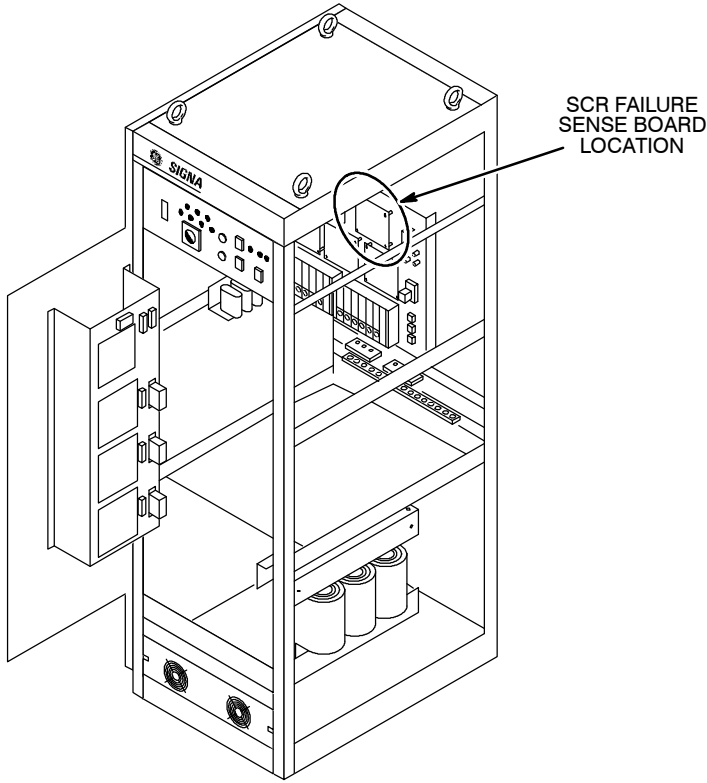
5--5 SCR FAILURE SENSE BOARD

PD1 A9 A2

PD1 A14 A2

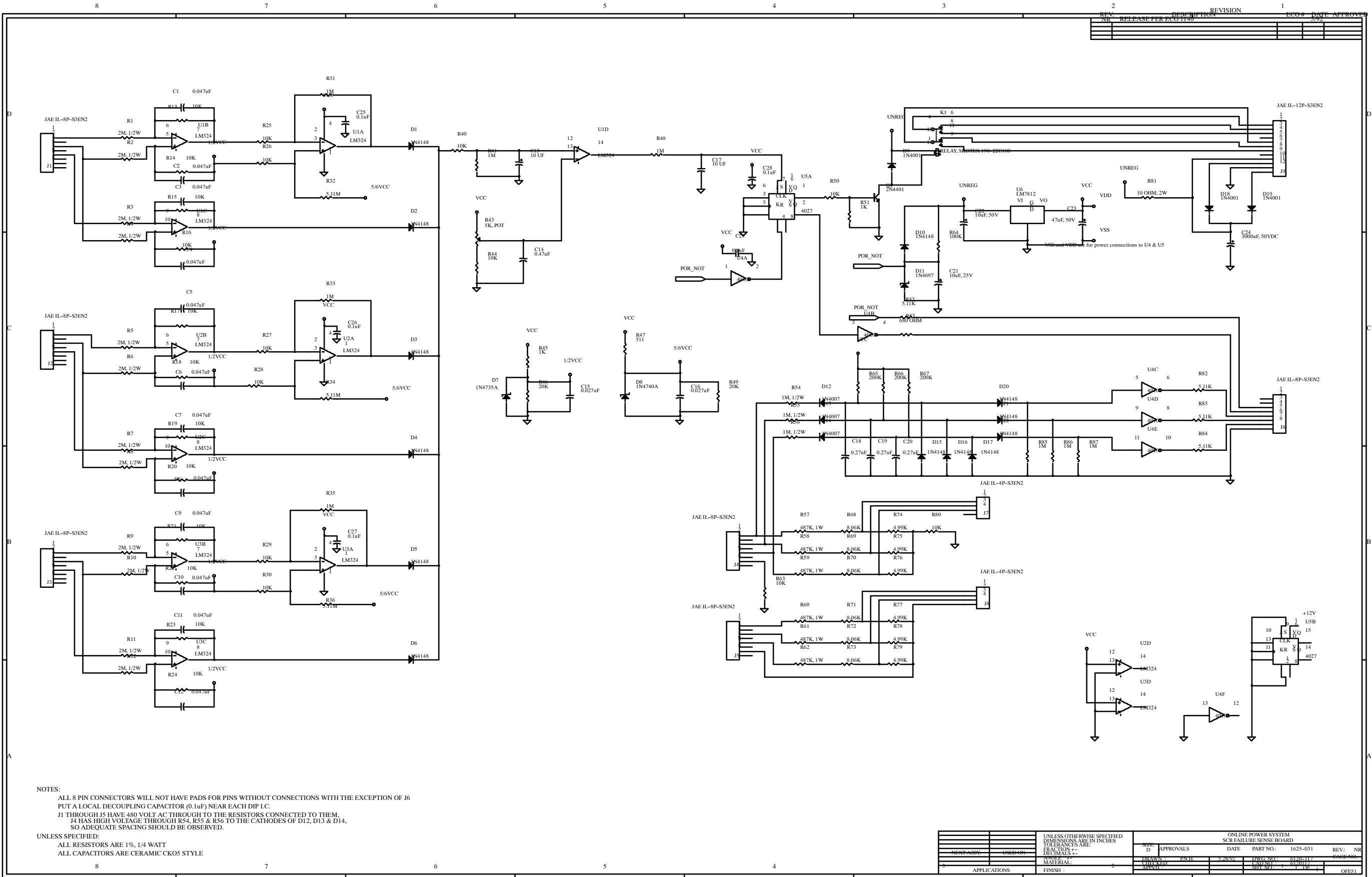
SCR FAILURE SENSE BOARD

46- -307819P15



Description

The SCR Failure Sense Board monitors fuselink continuity and will shunt trip the Main Circuit Breaker and provide a signal for the SCR FAILURE LIGHT if this circuit is open. Additional functions of the SCR Failure Sense Board are: providing input, output, and neutral signals to the front panel test points and providing 28Vdc power to the POWER OFF and EMER. STOP RESET switches and the interface board.

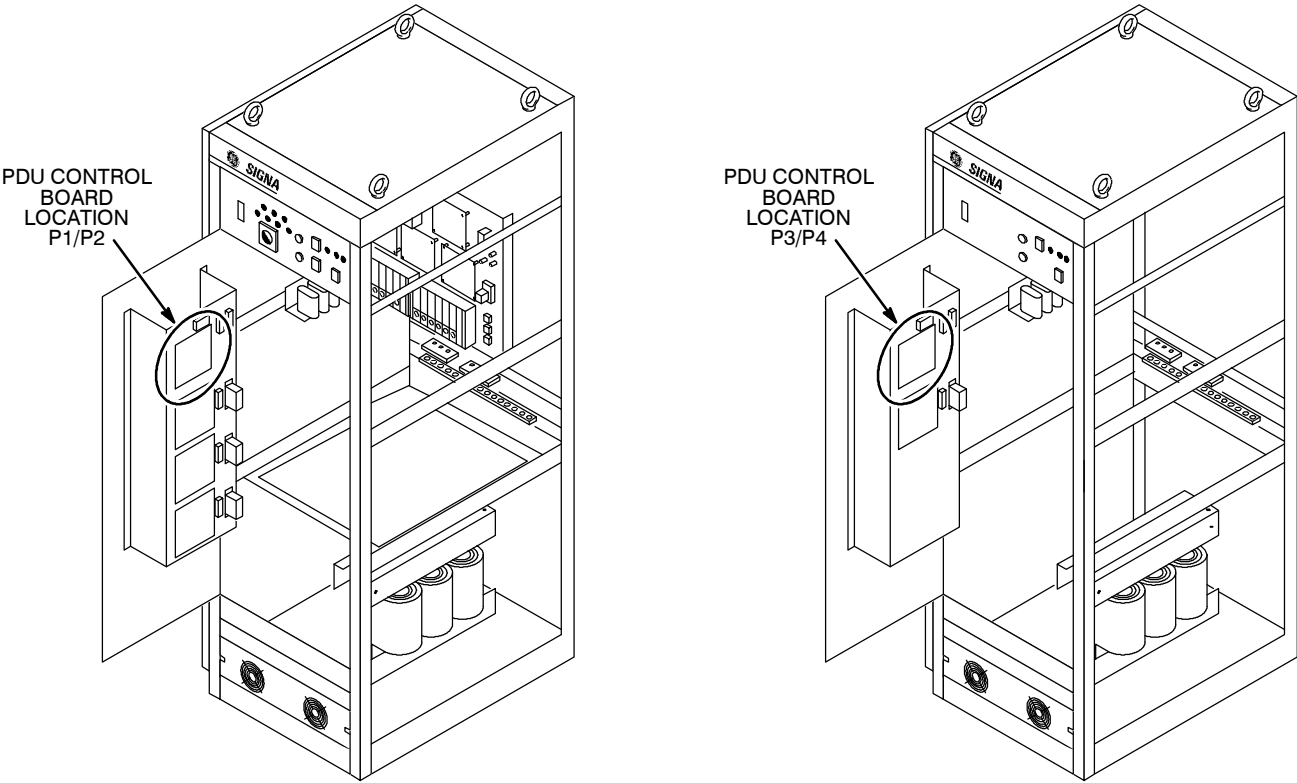


5--6 PDU CONTROL BOARD

PD1 A14 A1

PDU CONTROL BOARD

46- -307819P46

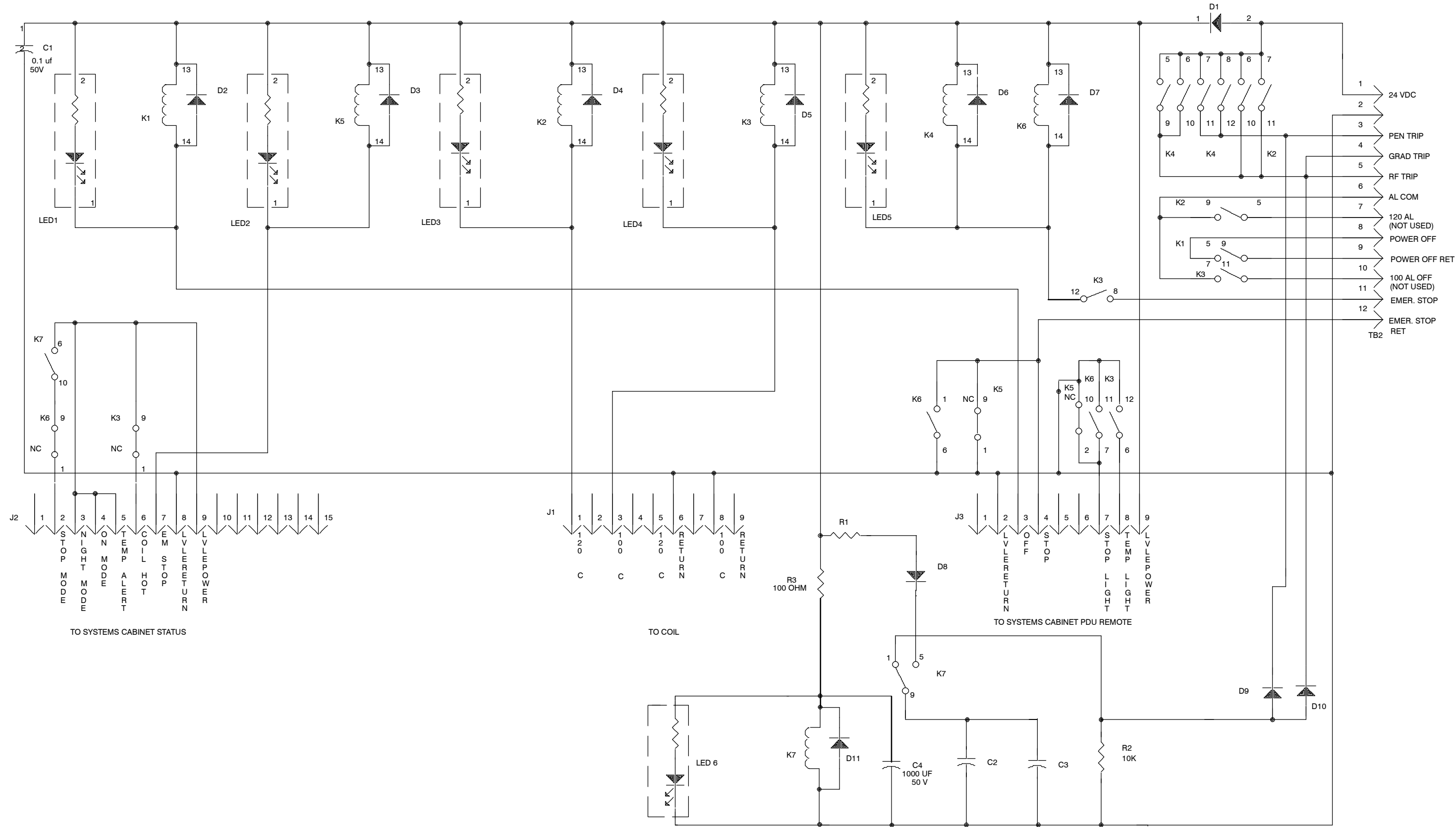


Description

The PDU Control Board allows Emergency Stop and overtemperature controls to shunt trip load panel circuit breakers. Activation of the Emergency Stop function from the Magnet Enclosure, Console, or the PDU Remote Control Panel located on the System Cabinet, trips the Gradient Amplifier Cabinet, RF Amplifier Cabinet, and Penetration Cabinet load panel circuit breakers. It also trips any option circuit breakers which have shunt trips, such as Broadband RF Amplifier Cabinet and Resistive Shim Power Supply. An overtemperature condition at the Compact PDU transformer or in the Systems Cabinet will shunt trip the Main Circuit Breaker. The connection points are J1 (transformer coil), J2 (System Cabinet status), and J3 (System Cabinet PDU Remote Control).

PDU CONTROL BOARD

PD1 A14A1

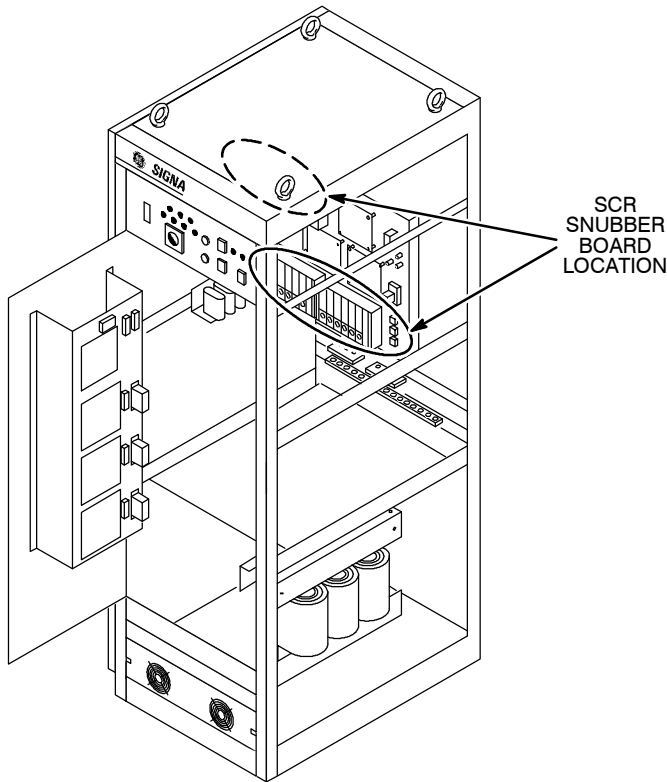


MFD/5CV
2200 MFD 35V

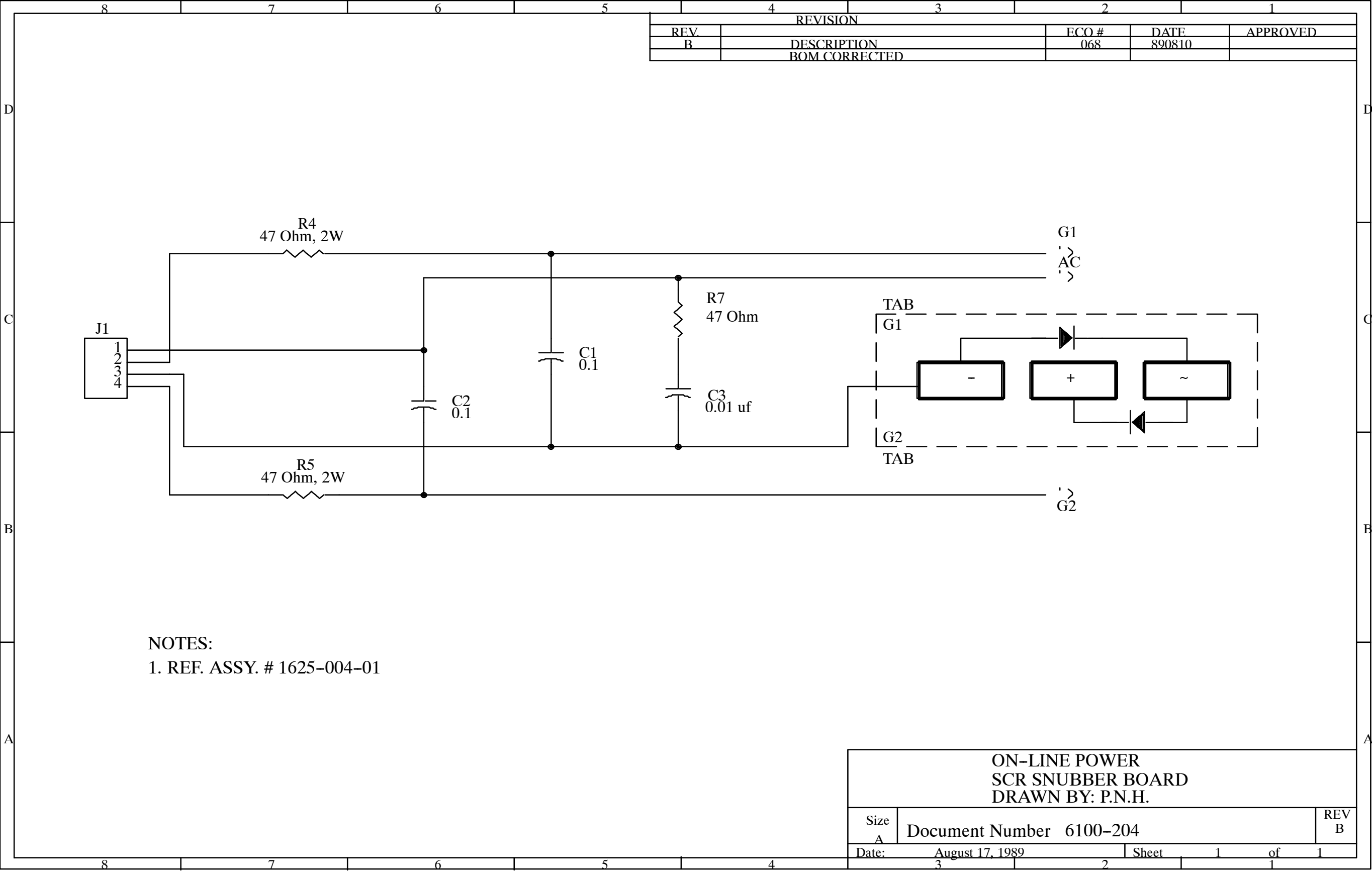


5--7 SCR SNUBBER BOARD

PD1 A9 A1, A6, A7
SCR SNUBBER BOARD
46- -307819P13

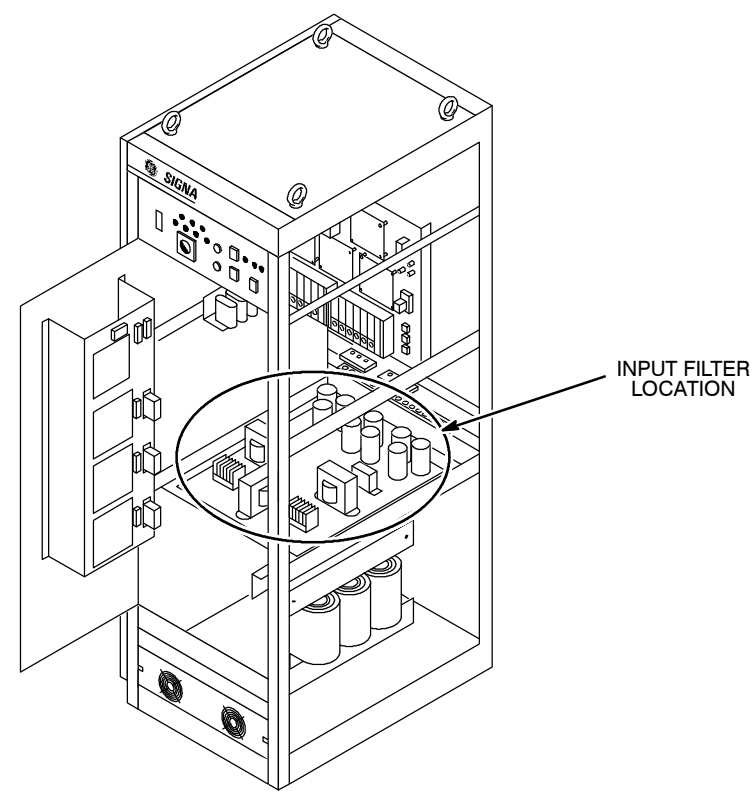


Description
The SCR Snubber Board applies the pulse transformer output to the SCR bank.



5--8 INPUT FILTER ASSEMBLY

PD1 A11
INPUT FILTER ASSEMBLY



Description
The Input Filter Assembly contains the Accuvar which removes peaks of voltage spikes, reduces the rate of rise of the voltage (a rapid rise in voltage damages SCRs), and removes high- -frequency noise from line power.

